

Application of Copyright Exceptions under Text and Data Mining Technology

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Abstract—In the era of big data, TDM technology plays an important role in scientific research, business, public health and other fields. However, the use of TDM technology still involves the risk of copyright infringement. When the legality of the use of TDM technology is unclear, because the data itself may be protected by copyright or sui generis right, TDM users need to pay a large amount of money. The cost and time of using TDM technology have greatly increased, hindering the application of TDM technology in various fields. Currently, China does not provide copyright exceptions for the use of TDM technology. Therefore, it is necessary to provide legal and technical guarantees for the development of TDM technology. Based on the legislative experience of the United States, Japan and the European Union, improvements can be made from the legal and technical aspects. At the legal level, TDM technology can be added to the existing fair use clauses, and impose restrictions on the purpose of technology use and the legality of data sources; At the technical level, a technical path for TDM copyright management and screening can be used to pre-screen data and exclude copyrighted data from the database, to reduce the risk of infringement of TDM technology.

Keywords—text and data mining- TDM , copyright infringement, copyright exceptions, technical path of text and data mining

I. INTRODUCTION

With the development of the data economy, the amount of data information is exploding every day. The OECD noted that far more digital information was produced between 2008 and 2011 than in recorded history[1]. When faced with massive amounts of information data, it is difficult to quickly and accurately find the required information. In this context, Text and Data Mining Technology emerged as the times required, helping people extract useful information from a large amount of information materials, which not only reduces the cost of manual data collection but also improves the efficiency and accuracy of data collection.

However, the use of TDM techniques needs to be supported by a large amount of data. Therefore, the use of TDM

techniques may face obstacles in four aspects: copyright directive, data base protection, contractual obligation and technical protection[2]. Here accurately point out the challenges faced by TDM from the legal and technical aspects, and the study will also be developed from these two perspectives. If no exception or limitation applies for TDM, the action may be prohibited by copyright or sui generis database right.

China has not made specific regulations on the application of TDM techniques, and its application in some specific fields will be subject to rights restrictions and may cause disputes. Therefore, this article focus on the possible copyright infringements involved in TDM techniques, and provide suggestions for improving the exceptions for China's TDM techniques based on an analysis of existing legal provisions in other countries.

II. THE BASIC THEORIES OF TEXT AND DATA MINING TECHNOLOGY

In the EU's DSM Directive, TDM is defined as "any automated analytical technique aimed at analyzing text and data in digital form in order to generate information which includes but is not limited to patterns, trends and correlations" [3]. Simply put, TDM can be regarded as an algorithm's process of extracting, analyzing, and evaluating data. But the definition of TDM in the DSM directive is too broad, and it can include any activity that uses digital technology to extract information. Through the analysis of technical principles below, it is find that data mining only exists in the Extracting stage, so data analysis can be better used to define TDM techniques. The principles of TMD technology are generally explained in the following three steps: Copying, Extracting and Recombining.

A. Copying

First, TDM needs to obtain information. At this stage, TDM uses mining algorithms to obtain or collect large-scale text and data sets, which can come from various sources such as the Internet, databases, scientific literature, social media, log files, etc. Then copy a large amount of content, such as text

documents, structured data tables, images, audio and video, etc. Most of the above information is obtained from relevant websites with data acquisition permission through crawler technology. Data that cannot be got on the Internet is supplemented manually and improved to provide a data basis for further subsequent data processing.

B. Extracting

After completing the collection of basic information, the machine will use it as a basis to establish models, or "patterns" [4]. In this process, since the acquired data usually contains noise, redundancy, and inconsistency, the data needs to be cleaned and preprocessed before text and data mining. This includes operations such as text segmentation, stop word removal, stemming, data standardization, missing value handling, etc. to make the data more suitable for analysis. After preprocessing, the text is converted into a numerical representation that the computer can understand. Such as converting original text, pictures or music materials into computer-recognizable symbolic language, or organizing and summarizing information, eliminating unnecessary information, and classifying useful information to facilitate subsequent cross-referencing of different data.

C. Recombining

This stage mainly involves reintegrating existing data and mining data sets to reveal trends, models, and correlations. Finally, for the purpose of TDM users, the data mining results are published. Data visualization tools and reports are usually used to effectively convey the mining results and findings to users, such as integrating the effective data in the form of documents and questions and answers to complete the entire process of data and text mining. Taking the establishment of a corpus model as an example, you first need to input a large amount of basic knowledge data into the machine. For example, establishing an intellectual property corpus requires inputting a large amount of information about intellectual property from web pages or manually into the machine, including but not limited to text and pictures. etc.; Secondly, the machine will convert all information materials into a computer language system, conduct machine learning after certain screening of the information, and integrate the data into the target data set required by the implementer. Finally, users of the corpus can ask the machine for the information they need to obtain, and the computer will extract relevant data from the target data set, organize and transform the language, and output the mining results to complete the entire TDM process. These three steps will be illustrated by the following figure.

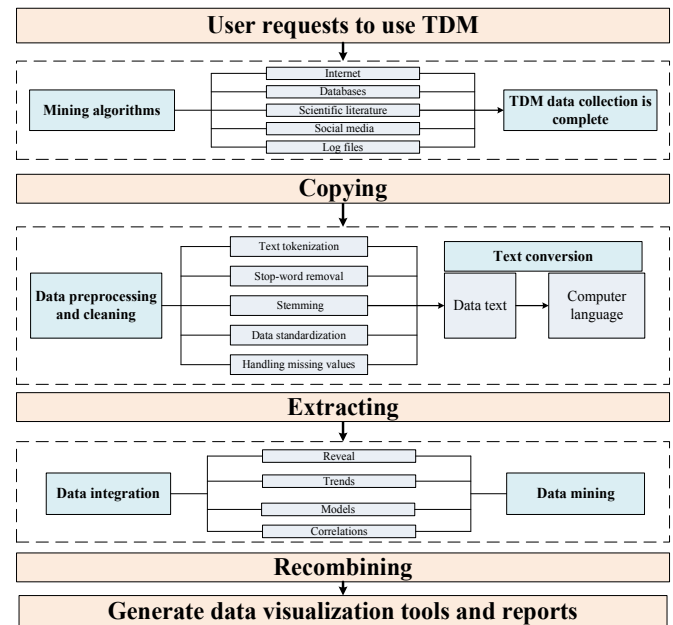


Fig. 1. TDM technology path.

III. COPYRIGHT INFRINGEMENT RISKS OF TDM TECHNIQUES

In the process of implementing TDM techniques, the collection of original data will inevitably be involved. For data and materials that enjoy copyright, TDM may infringe the right of reproduction, adaptation, and information network dissemination; for works that do not enjoy copyright, the mined data and text may also be protected by sui generis right.

A. Copyright

In the copying process, TDM may infringe the right of reproduction of the work. Before analyzing text or data, the machine first needs to make a copy. According to the idea/expression dichotomy, copyright only protects original expressions, not ideas, principles, data, etc. However, even if the content is not protected by copyright, the container may be controlled by rights. In other words, although the data and text may not enjoy copyright, the database composed of data and text may be protected by copyright. Therefore, when exploiting data and text, rights protection also extends from the container to the content[5]. When TDM techniques is used to extract and copy copyrighted databases and convert them into machine-readable language, the right of reproduction may be infringed. During the Extracting stage, TDM may infringe upon the adaptation or translation rights of the work. At this stage, the computer mainly performs language conversion on the collected work information, converting the content of the work into a language that can be recognized by the computer, such as converting a PDF document into an XML file. The act of transcoding itself does not involve a change in the expression of the original work, but is merely a conversion of the language system, similar to a special act of translation, so this stage may fall under the protection of translation rights. At the same time, Extracting will also delete and filter useless information, changing the expression of the original work, which may infringe the right of adaptation[6]. In the final Recombining stage, TDM techniques will output the data mining results to

the public. In the digital environment, authors are also given the right to communicate to the public of the work[7]. If the information obtained by the public is protected by copyright, it is difficult to avoid the risk of infringement of the right of information network dissemination.

B. sui generis right

Under the TRIPS Agreement, creative databases can be protected by copyright, and sui generis rights can be used to protect creative and non-creative databases. Most databases do not meet the originality requirements in the sense of copyright law, so the introduction of this right can fill the gap of legal protection and stimulate the dissemination and acquisition of information[8]. Sui generis right is similar to copyright, both are exclusionary rights, but they have different judgment methods: the core of copyright protection lies in whether the database is originality, and whether sui generis right is enjoyed is based on the acquisition and construction of the database involve labor costs. Only when the owner of the database proactively takes measures to deal with the risk of infringement and invests a lot of manpower and material resources in the database can he claim ownership rights to the database. For example, The EU Court of Justice made a judgment in 2004 that when the investment is used for the obtaining, presentation and verification of the database, sui generis right can be cited to protect the database; and when the investment is used for the creation of data, then are no longer protected by this right to reduce the risk of abuses of dominant positions[9]. When the maker of a database has substantial investment in the obtaining, verification or presentation of the database, it has the right to prevent others from extracting or reusing the entire or substantial content of the database[10].

IV. THE EXISTING TDM COPYRIGHT EXCEPTION SYSTEM

In order to meet the demand for the use of TDM techniques in the big data era, various countries have stipulated the scope of fair use of TDM techniques through different legal provisions to ease the tension between technology and intellectual property rights.

A. The US model

The United States has not passed a clear legal provision for copyright exceptions to TDM, but has recognized the application of the fair-use doctrine to TDM techniques through judicial precedents. The judgment of fair use under U.S. copyright law is mainly divided into four steps, namely the purpose and character of the use, the nature of the copyrighted work, the amount and substantiality of the used part in the overall work, and the impact of the use behavior on the market value of the work[11]. In summary, the United States adopts a relatively flexible judgment method for the reasonable use of TDM, and does not use commercial use as a basis for judgment. Its main purpose is to promote the development of TDM techniques and give full play to its role in the commercial field, so as to bring greater economic value. But at the same time, non-prohibitive regulations on commercial use will cause commercial entities to use TDM techniques to compete for the market and transfer the interests that belong to the copyright owner to the hands of the company.

B. Japanese model

Japan passed the revision of the Copyright Law in 2018, and stipulates in Article 30, paragraph 4, that when using a work in the process of information analysis or electronic computer processing of information, if it is not for the purpose to personally enjoy or cause another person to enjoy the thoughts or sentiments expressed in the work, it does not constitute copyright infringement[12]. TDM techniques mainly analyzes data and can fall into the situations stipulated in Article 30-4 of the Japanese Copyright Law; and the use of TDM techniques for works does not involve the analysis of the thoughts and feelings of the works, but only the extraction of effective information, therefore constitutes an exception to copyright. But when it comes to commercial use, Japanese copyright law prohibits it.

C. EU model

In March 2019, the European Parliament reviewed and approved Directive on Copyright in the Digital Singles Market, and stipulated copyright exceptions for TDM in Articles 3 and 4. According to the provisions of Article 3, research organizations and cultural heritage institutions may copy and extract TDM of legally acquired works for the purpose of scientific research, and the research is limited to not-for-profit and with a public interest objective[13]. Article 4 stipulates that member states should allow the copying and extraction of legally obtained works or other content for the purpose of text and data mining. Therefore, copyright exceptions can only apply when the user has had lawful access to the content. These two exceptions are widely called for by the scientific community, because TDM is an important tool to promote innovation in the era of big data[14]. This will improve the legitimacy of TDM's use of data and reduce the risk of infringement from the source. Like Japan, EU law also prohibits the commercial use of TDM.

TABLE I. COMPARISON OF SYSTEMS ACROSS COUNTRIES

Comparison of TDM copyright exception systems			
Country	Statute	Purpose	Commercial use
The US	Through judicial precedents	Transformative use	Allow
Japan	Copyright Law in 2018	Not for the purpose to personally enjoy	Ban
EU model	Directive on Copyright in the Digital Singles Market	For the purpose of scientific research	Ban

V. METHODS ON INTEGRATING TDM TECHNIQUES INTO CHINA'S COPYRIGHT EXCEPTION SYSTEM

Chinese Copyright law do not provide copyright exceptions for the application of TDM techniques, so there is legal uncertainty when using TDM techniques. Based on the huge

role and value of TDM techniques in the big data era, it is necessary to solve the legality issue of application. Therefore, we can start from the legal and technical levels to achieve a balance between TDM techniques and intellectual property protection.

A. Legislative discussion: Introducing TDM fair use clauses

Currently, there are three main ways to restrict copyright: implied license, legal licensing and fair use. Implied permission means that under certain circumstances, the right holder's silence can be presumed as an indication of permission to use. Although the implied license simplifies the transaction procedure, the application of TDM techniques has not yet formed a market practice, so the applicability of the implied license is not high; legal licensing requires the law to clearly provide for use exceptions in advance. On the one hand, there is the problem of high legislative costs. On the other hand, it is also difficult to cope with the adjustments and changes in the digital market. Therefore, China can choose the doctrine of fair use as the path to TDM copyright exceptions, and add TDM fair use clauses on the basis of the current copyright law that stipulates 12 specific situations of fair use.

First of all, in terms of the subject setting of the fair use of TDM, we can learn from the EU regulations as "scientific research institutions and public organizations"; on this basis, other institutions or enterprises should also be allowed to conduct TDM research for non-commercial purposes. For example, the 2016 amendment to France's Intellectual Property Code stipulates that the public and researchers can mine legally acquired works for non-profit scientific research purposes[15], to fully utilize the potential value of data. Taking medical institutions as an example, if the initial purpose of engaging in TDM research is to conduct scientific research or conduct teaching activities, and the original intention of the research is non-commercial and is beneficial to the promotion of public interests, it can be included in the scope of fair use of TDM.

Secondly, in addition to scientific research purposes, the purpose of fair use of TDM can also be expanded to other fields, and the "three-step test" can be used to check whether it complies with fair use behavior. At present, TDM techniques has played a huge role in language research and Medical Science, and it may also be used for data mining and trend analysis in other fields in the future [16]. Therefore, as long as the use of TDM does not constitute infringement, does not damage the rights of others' works, and does not affect the normal use of other people's works, it can be regarded as fair use of TDM. The expansion of applicable fields can allow TDM to play a greater role in research and life, so as to promote the development of TDM technique. For commercial purposes, TDM fair use provisions are temporarily not applicable. On the one hand, taking into account the uniformity of the law, the existing fair use provisions in China's Copyright Law do not include commercial activities. Therefore, it is not appropriate to set exceptions for TDM; on the other hand, if the exception for fair use of TDM commercial behaviors is opened, due to the large demand of the commercial market, and without corresponding provision in other policies, large-scale open data mining activities will lead

to the legal rights of data rights holders are harmed, which violates the purpose of the copyright law.

Finally, the fair use of TDM should be based on the of data and text resources that legally access. In the EU's Copyright Directive, there are restrictions on "lawful acquisition" of mined works; similarly, "lawful acquisition of works" is also the prerequisite for the British Copyright Act to allow actors to conduct TDM. In response to this regulation, some scholars have raised questions: Large institutions in the market have legal access to a large number of works based on their abundant resources, while small institutions only have limited resources. In addition, the difference in data mining capabilities between the two will lead to small institutions lose their market competitiveness[17]. However, the prerequisite of "legal acquisition" should still be retained in China's legislative settings to prevent actors from conducting TDM through illegal means, such as undermining technical measures, thus damaging the legitimate interests of copyright owners. For competition among research institutions of different sizes, the Anti-Unfair Competition Law can be used to regulate it. At the same time, the legality of the data source can also ensure the accuracy of the data itself, then to ensure the reliability of TDM technology execution.

B. A technical path idea for TDM copyright management and screening

Since the implementation of TDM technology requires copying and extracting a large amount of data and text, and outputting the results, there is a risk of copyright infringement at every step of the process. At present, most disputes related to TDM are ex post relief measures, that is, corresponding measures are taken after the infringement results have occurred. However, this will cost the rights holder a lot of money and time, and will also hinder the use of TDM technology in subsequent research. Therefore, the data preprocessing and cleaning stage in the data text mining process can be optimized, and a copyright screening and authorization management system can be introduced to filter copyright-protected data to reduce the risk of copyright infringement caused by data text mining :

First, a basic copyright database needs to be built, which contains data and texts currently protected by copyright. This database is collected and provided by publishers, authors and copyright collective management organizations.

Secondly, after the data processing and cleaning steps, perform fingerprint analysis on the data and calculate it through a hash function so that each data text corresponds to a unique identification code. These data texts marked with identification codes are then compared with the metadata related to copyright statements or intellectual property rights extracted from this basic copyright database.

Finally, regular expressions are used to match specific copyright claims and reference databases are used to detect references to known works to identify patterns in copyright text to determine whether the data is protected by copyright.

After determining that a text data item is protected by copyright, the system will use information extraction technology, such as named entity recognition (NER) to identify

and extract the copyright statement, author name, publication year and other information of the data text based on the self-built authorization database, and then use text matching algorithms to identify copyrighted content and then determine its authorization status.

If the data text has been authorized, proceed to the next step after recording. For unauthorized data and text, the system will select them, and the following operations can be performed based on user selection: 1. Directly remove this data text; 2. Automatically generate an authorization request and send it to the relevant copyright holder or collective management organization; 3. Provide alternative data sources, such as open access resources or materials whose copyright has expired.

After passing through the copyright screening and authorization management system, these authorized or open source data texts are recorded and compliance reports are generated to prepare for audits and legal needs. Then enter the data integration process and finally generate a visual data report. Using the above technical paths to perform copyright screening on the data used by TDM can reduce the risk of TDM technical copyright infringement from the root cause. While using the law to protect the legality of TDM technology, it can also provide a technical protection path for copyright.

This solution emphasizes specific copyright-related pre-processing steps to reduce copyright risks. At the same time, existing TDM technology does not pay attention to the compliance and legality of its technical means, but this plan emphasizes the importance of generating compliance reports through a systematic process to legitimize it. The user interaction method of this system can make data processing more flexible and better meet the needs of users. However, general TDM technology may less consider the active participation of users, and data processing may be more automated, while users are faced with copyright issues, may not have the flexibility to choose how to process the data. The systematic application of this technology and the degree of user participation in copyright management are superior to traditional TDM technology. This comprehensive processing method makes TDM technology applications more reliable and compliant, reduces potential legal risks, and provides users with more flexible and trustworthy data processing solutions.

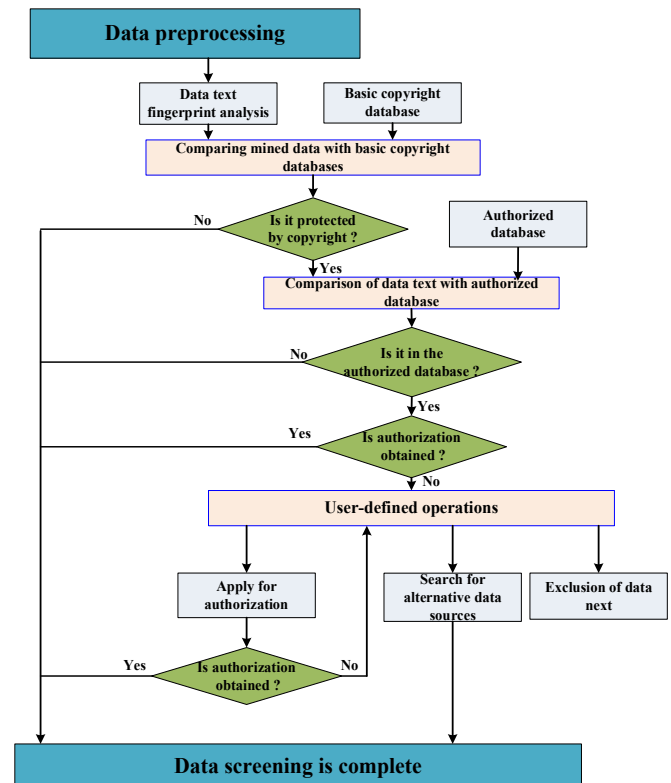


Fig.2. technical path for TDM copyright management and screening.

VI. CONCLUSIONS

While the development of new technologies brings convenience to work and life, it also causes legal uncertainty. The development of TDM technology is one of them. This article starts with the principles of TDM technology, analyzes the obstacles that existing copyright law brings to the development of TDM technology, and makes suggestions for the development of TDM technology in China from the perspective of comparative law. Under China's copyright law system, the use of TDM technology still carries a large risk of infringement. Therefore, it is necessary to provide protection for the development of TDM technology from both legal and technical levels. The law mainly provides support for the legality of TDM technology. Judging from the current legislation, the choice of the principle of fair use is the most feasible, so the existing fair use provisions can be revised or interpreted to fit it. The use of technology is to screen data and texts in advance to avoid possible infringement risks for TDM technology. Therefore, the law should adapt to the development of new technologies, and technology should also be applied within the legal framework to achieve a balance between technology and law, while protecting the interests of rights holders and bringing more impetus to social development.

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